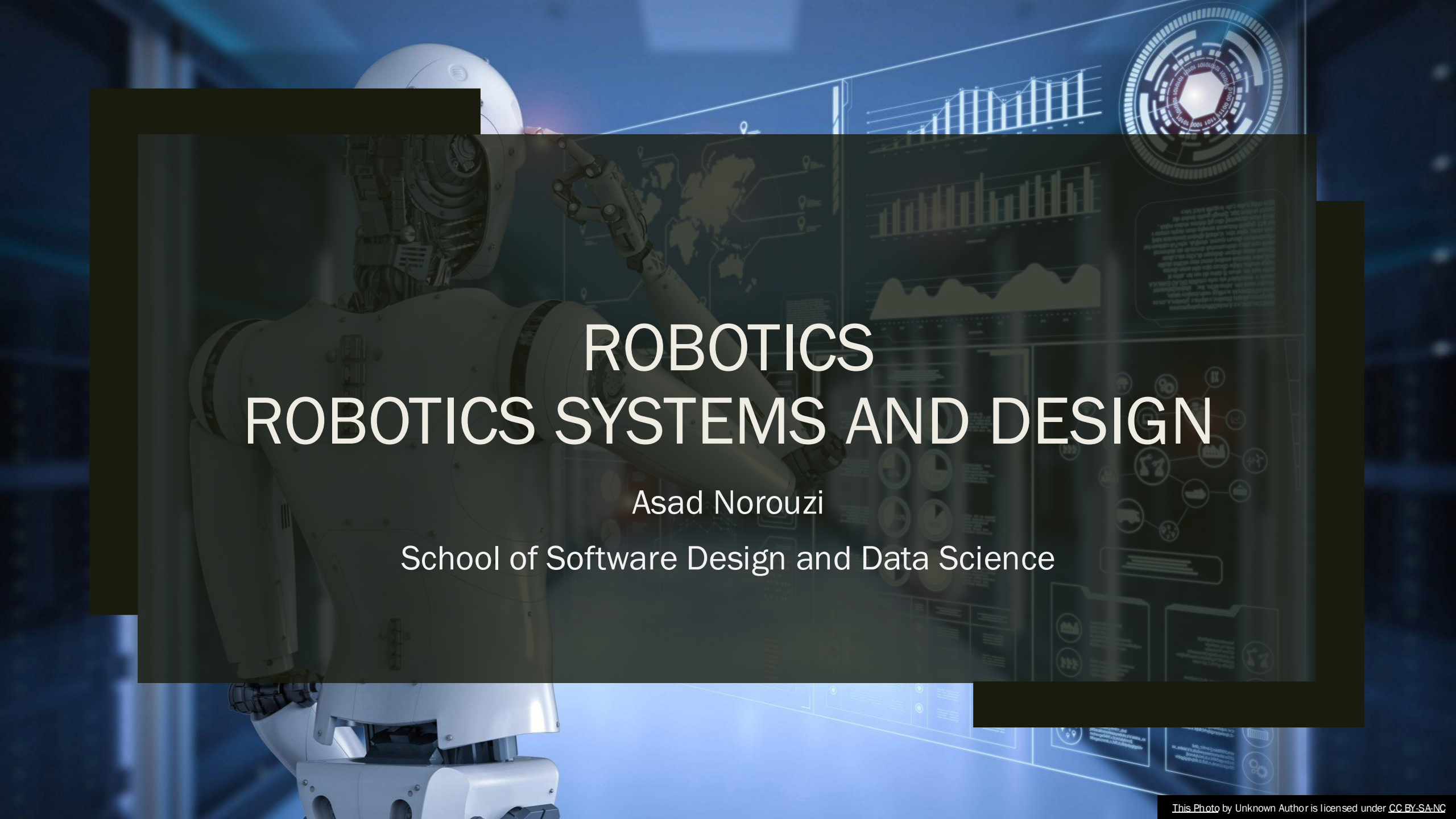


A white humanoid robot is shown from the side, interacting with a large, futuristic digital interface. The interface displays various data visualizations, including bar charts, line graphs, a world map, and a circular gauge. The robot's right arm is raised, with its hand touching the screen. The background is a dark blue, slightly blurred, suggesting a high-tech environment.

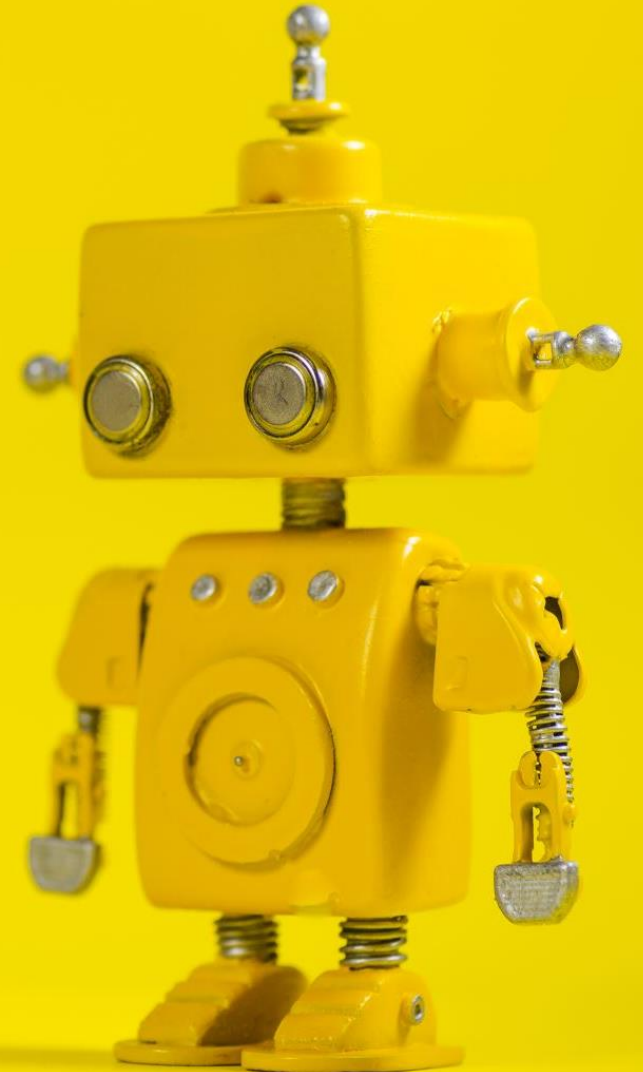
ROBOTICS ROBOTICS SYSTEMS AND DESIGN

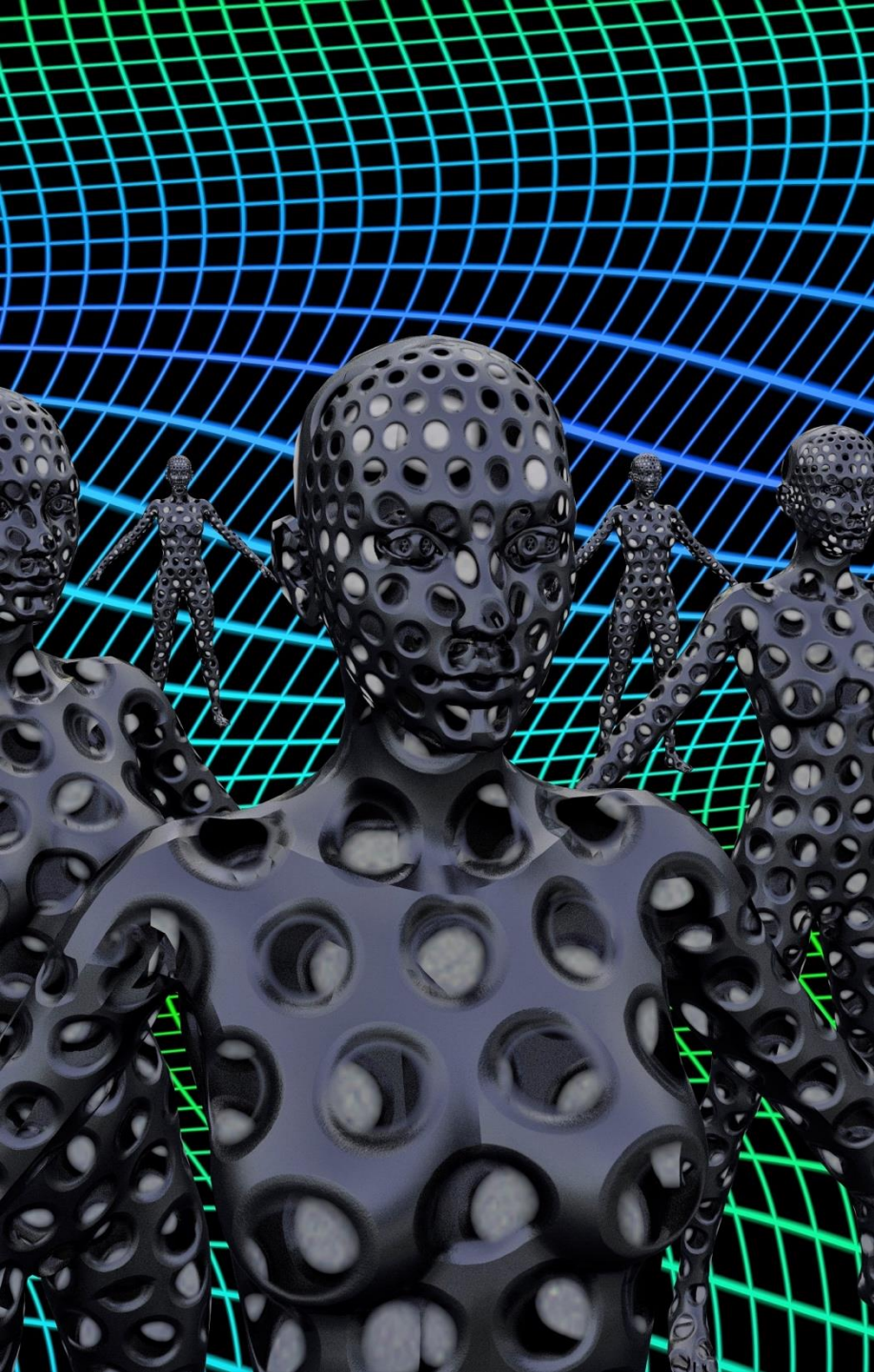
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Agenda

- Overview of Robotics Systems
- Mechanical Design
- Electrical and Electronic Systems
- Sensors and Perception
- Control Systems
- Software Architecture
- Integration and System Design
- Summary
- Q&A





Overview of Robotic System

A robotic system is an integrated set of hardware and software components designed to perform tasks autonomously or semi-autonomously.

Mechanical Structure: The physical form and movement mechanisms.

Sensors: Devices that collect data about the environment and the robot's own state.

Actuators: Components that produce movement or other physical effects.

Controllers: Systems that manage the robot's actions based on sensor inputs.

Software: Algorithms and programs that dictate the robot's behavior.



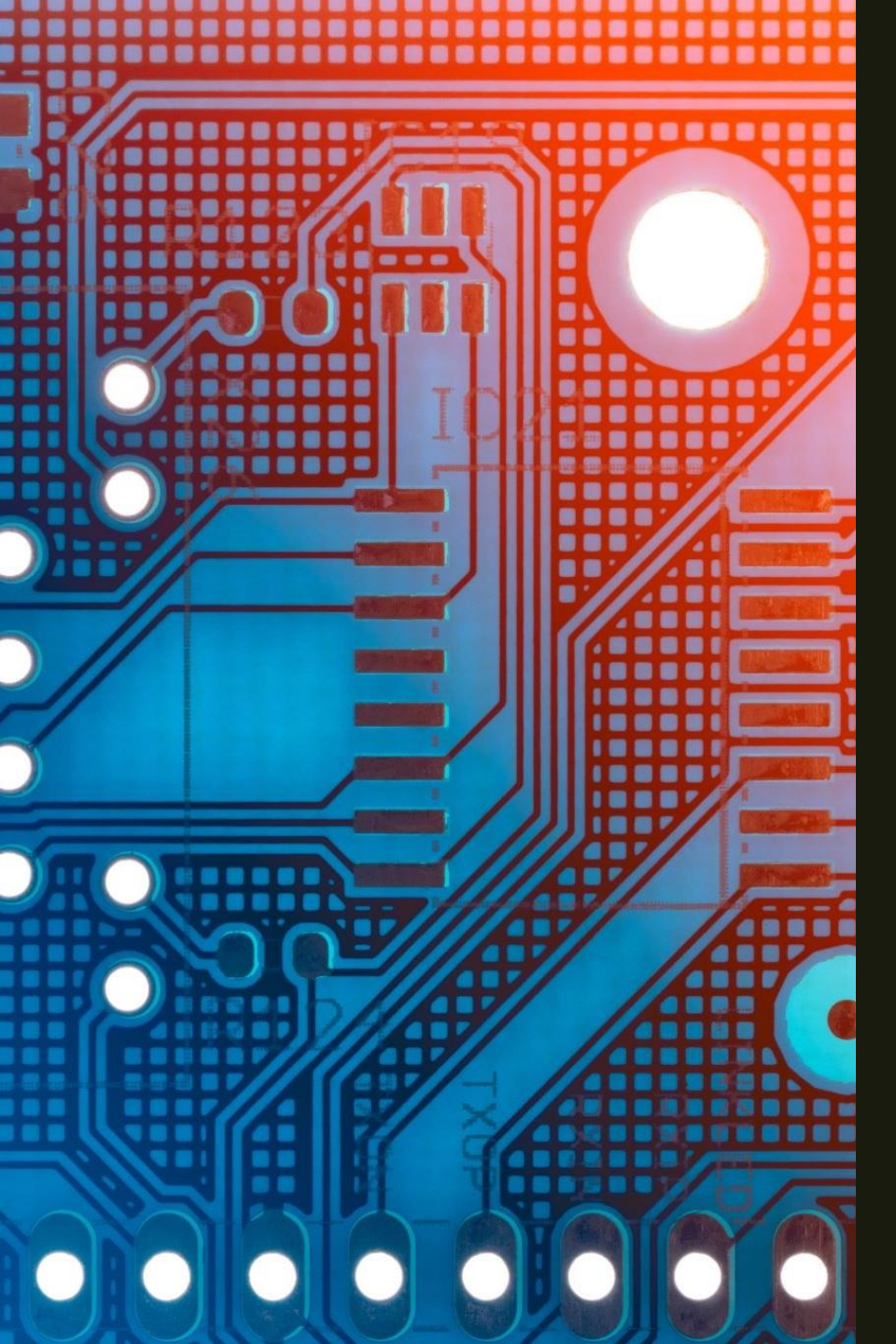
Mechanical Design

Kinematics and Dynamics: Study of motion (kinematics) and forces (dynamics) that affect the robot's movements.

Degrees of Freedom (DOF): Number of independent movements a robot can perform.

Material Selection: Choosing appropriate materials for strength, weight, and durability.

End Effectors: Tools attached to the end of a robotic arm to interact with the environment (e.g., grippers, welding torches).



Electrical and Electronic Systems

Power Supply: Options for powering robots, including batteries and wired power.

Motors and Actuators: Types of motors (DC, stepper, servo) and their control mechanisms.

Control Electronics: Microcontrollers and FPGAs for real-time control and data processing.

Communication Systems: Wired (e.g., CAN bus) and wireless (e.g., Wi-Fi, Bluetooth) communication technologies for robot coordination and control.



Sensors and Perception

Vision Sensors: Cameras, depth sensors, LIDAR.

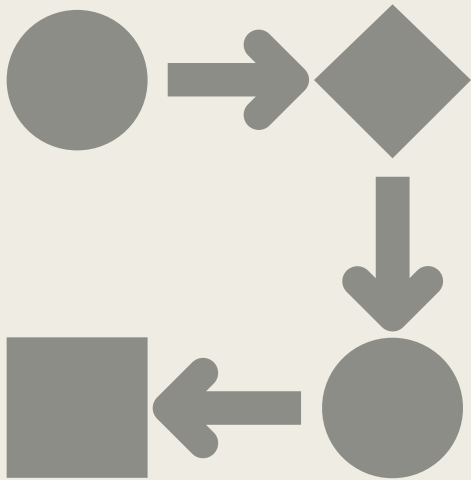
Proximity Sensors: Ultrasonic, infrared.

Inertial Sensors: Accelerometers, gyroscopes.

Force and Tactile Sensors: Strain gauges, pressure sensors.

Sensor Fusion: Combining data from multiple sensors to improve accuracy and reliability.

Control Systems

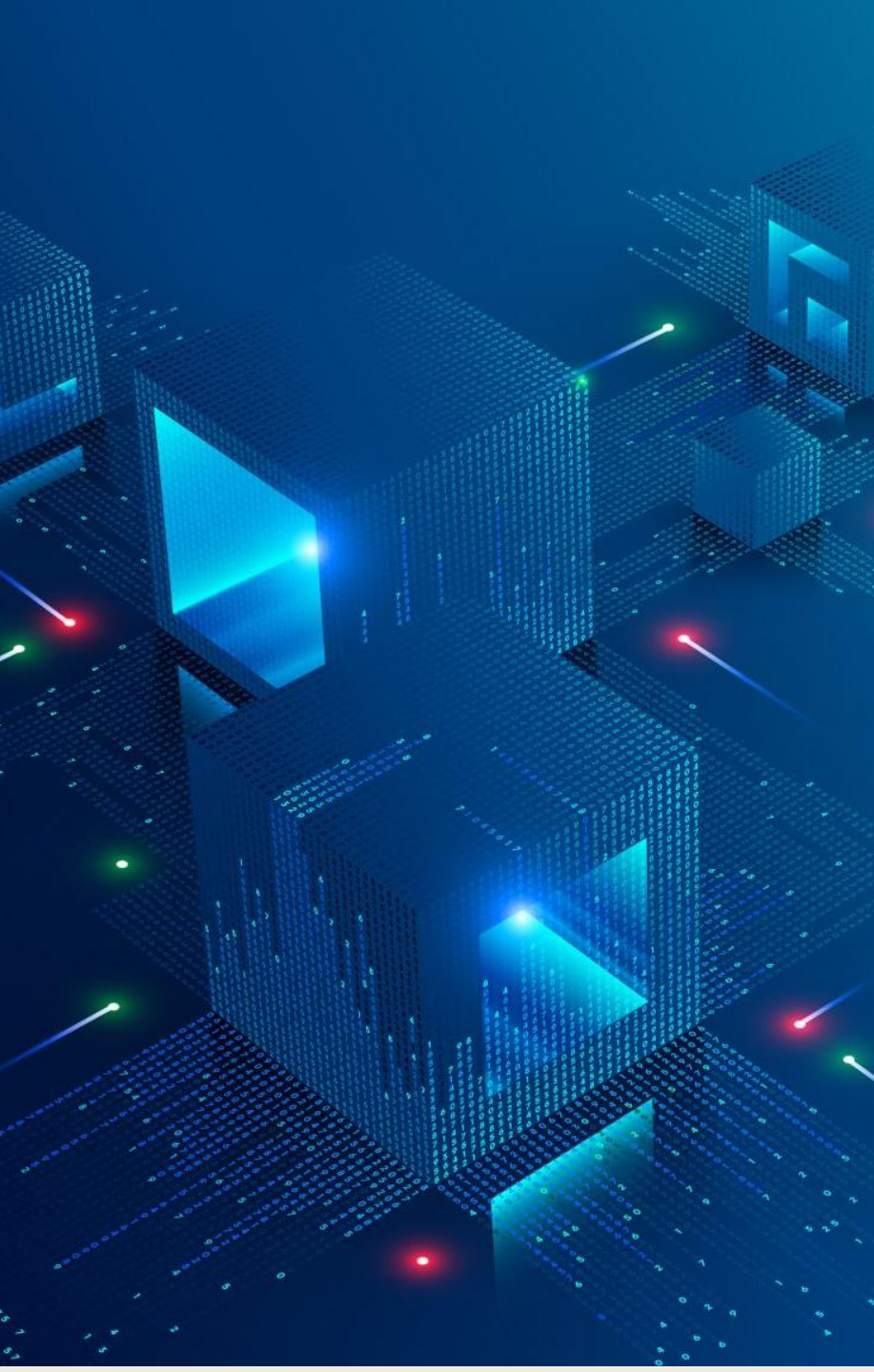


Open-Loop Control: A type of control system that acts solely based on the input, without using feedback to adjust its action.

Closed-Loop Control: A type of control system that uses feedback from the output to adjust its actions to achieve the desired outcome.

PID Control: Proportional-Integral-Derivative control for maintaining desired outputs.

Advanced Control Methods: Model predictive control, adaptive control, and robust control.



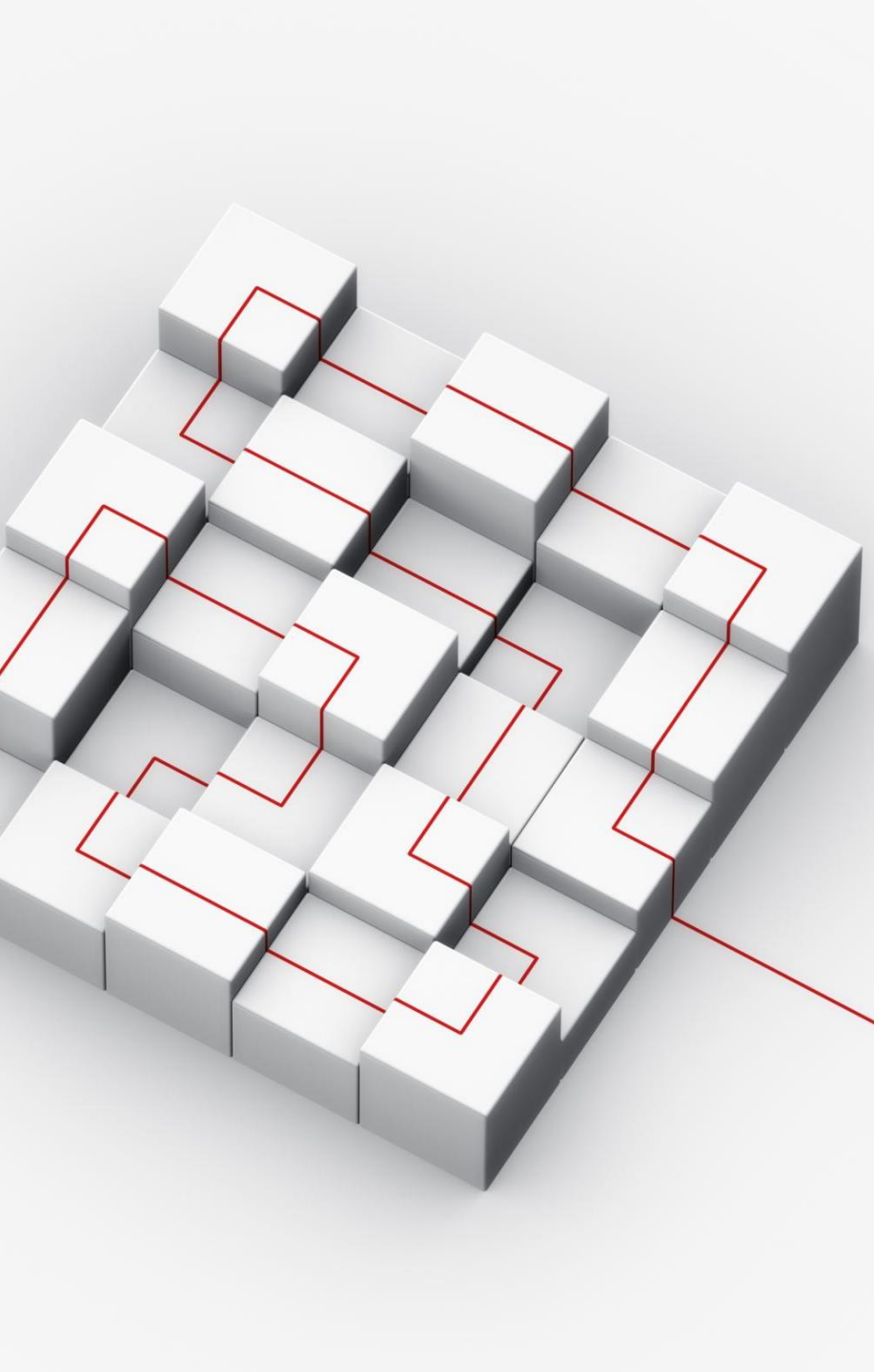
Software Architecture

Robot Operating System (ROS): A flexible framework for writing robot software.

Middleware: Communication protocols and data handling between software components.

Behavioral Programming: Finite State Machines (FSMs) and behavior trees for managing complex robot behaviors.

Simulation and Testing: Use of simulators like Gazebo to test robot designs before deployment.



Integration and System Design

System Integration: Combining mechanical, electrical, and software components into a cohesive system.

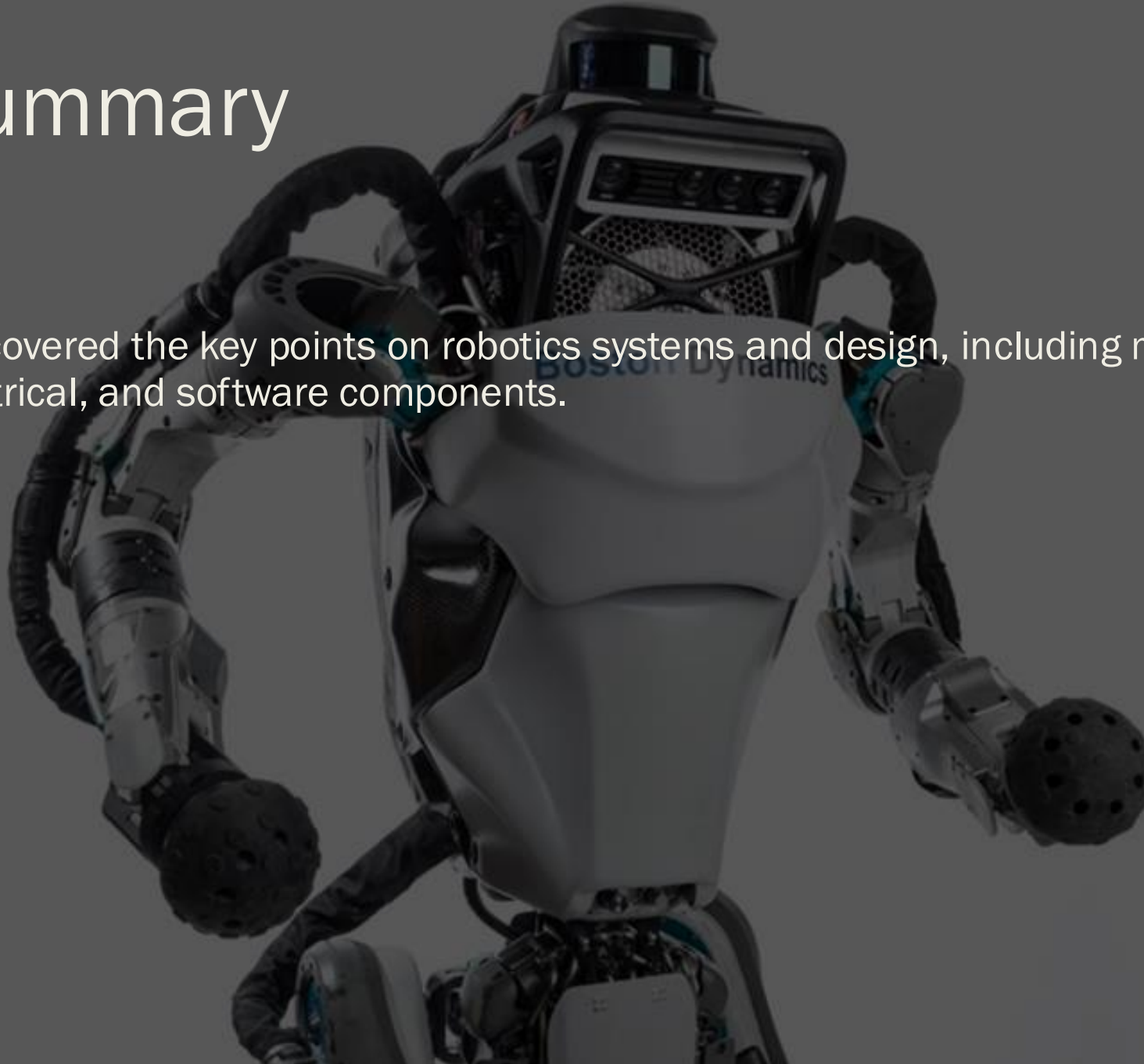
Design Methodology: Iterative design, testing, and refinement process.

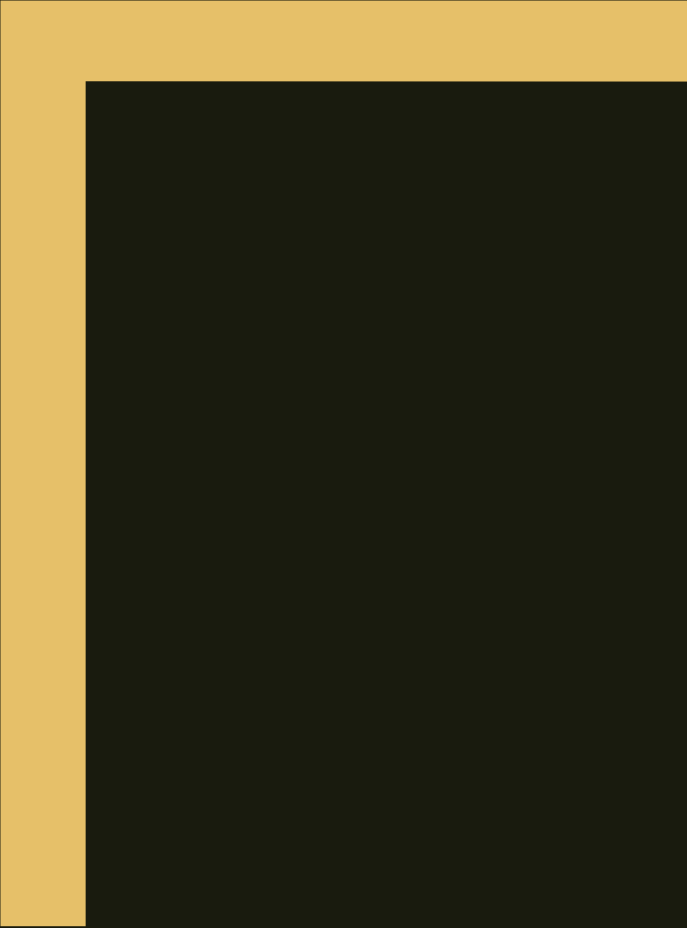
Prototyping: Creating prototypes to validate designs and functionality.

Modular Design: Designing robots with interchangeable and upgradable parts.

Summary

We covered the key points on robotics systems and design, including mechanical, electrical, and software components.





Q&A